

VTech AI Solutions Business Website Platform Requirements Specification

Compiled for Vithusan V • March 01, 2026

(React + Spring Boot + MySQL • CMS + Admin Panel + Leads + Collaboration)

Purpose

Define the complete functional and non-functional requirements for a production-ready business website platform. The platform must allow VTech AI Solutions to showcase services and projects, capture leads, and let an admin manage all public-facing content dynamically from an admin panel (including front page editing), without code changes or hardcoded content.

High-level goals

- Professional public website to communicate with clients and convert visitors into inquiries.
- Full CMS capability: all landing page sections, buttons, links, and content must be editable via admin UI.
- Admin workflow: draft, preview, schedule, publish, unpublish, archive; with visibility toggles and time windows.
- Two project types (Campus vs Business) with strong project cards/templates and direct links to detail pages.
- Lead capture: project requests, contact/WhatsApp, and collaboration requests; with admin inboxes and status pipelines.
- No data loss: soft delete for all entities and a Recycle Bin to restore or permanently delete.
- Auditability: user activity logs + admin audit logs.
- User accounts: register/login for profile features (wishlist/saved projects; optional payment card token storage).
- Responsive, user-friendly UX for mobile/tablet/laptop.

1. Technology Stack & Architecture

Required stack

- Frontend: React (Vite), professional theme, responsive design.
- Backend: Spring Boot REST API.
- Database: MySQL (all content and configuration stored in DB).
- Authentication: JWT with role-based access (ADMIN, USER; optional EDITOR).
- File uploads: images and videos for projects, ads, and page sections.

Architecture principles

- No hardcoded content for public pages: render from database-driven page sections and content JSON.
- Template-based CMS sections (pre-built professional layouts) rather than free-form HTML.
- Consistent global visibility/scheduling engine applied to all content types.
- Soft-delete everywhere; never permanently delete outside Recycle Bin.
- Strict validation, secure uploads, and audit logging.

2. Global Rules (Applies to Every Module)

- No hardcoding
 - All public-facing text, buttons, links, social icons, page sections, and displayed cards must be stored in DB.
 - Admin must be able to adjust content from UI after initial deployment.
- Publishing workflow
 - Statuses: DRAFT, SCHEDULED, PUBLISHED, ARCHIVED.
 - Admin can preview before publishing (token-protected preview).
 - Admin can publish now, schedule publish, unpublish, and archive.
- Visibility toggles & scheduling
 - Every entity supports: isVisible (toggle), visibleFrom, visibleUntil, publishedAt, createdAt, updatedAt.
 - Frontend only renders content if isVisible=true and current time is within visible window for PUBLISHED/SCHEDULED items.
- No data loss
 - Soft delete for all entities (deleted flag + deletedAt + deletedBy).
 - Recycle Bin UI to restore or permanently delete (with warnings/confirmations).
- Logging
 - Store user activity logs and admin audit logs in DB with filters and pagination.
- Responsive UX
 - Mobile-first layouts for public site and admin panel (collapsible sidebar, usable tables, consistent spacing).

Admin access link placement

The public website must show the Admin link only at the bottom of the page (footer). Suggested placement: small 'Admin' link under Footer > Resources. Public users should not see admin controls.

Default Admin Account (Seeded)

Seed an initial admin account in the database:

Username: VTN02V

Password: 02VTN@2#

- Passwords must be BCrypt-hashed in DB.
- Do not hardcode the password in controllers; seed via secure bootstrap (env vars allowed).
- Force password change on first admin login.

3. Dynamic CMS & Front Page Builder

CMS requirements

- Home page and other public pages must be built from DB-driven sections (page_sections + section content JSON).
- Admin can add new sections from templates (HERO, SERVICES_GRID, FEATURED_PROJECTS, TESTIMONIALS, UPDATES, FAQ, CTA_BANNER, SOCIAL_BAR, FOOTER, etc.).
- Admin can reorder sections (drag-drop or up/down), hide/show, schedule visibility, and preview changes.
- All CTA buttons must be configurable: label, target URL, style variant, and visibility.

Admin 'Edit Mode' on Front Page

When an admin is logged in, the public front page must support an overlay edit mode so the admin can manage key elements directly from the front page (using buttons and inputs) without editing code.

- Floating 'Edit Page' button appears only for admins.
- Inline controls per section: Edit, Add Section, Reorder, Hide/Show, Delete (moves to Recycle Bin).
- Save Draft and Preview changes; publishing should remain explicit (Publish Now / Schedule).

Updates / Announcements on Landing Page

- Admin can create site updates/announcements for users and landing page (title, message, optional image).
- Placement options: landing section, top bar, banner, or popup; with scheduling and visibility toggles.

4. Projects Module (Campus & Business)

Project types

- Every project must have a type: CAMPUS or BUSINESS.
- Public UI provides filters/tabs: All, Business Projects, Campus Projects.
- Project cards must display a clear badge for the type.

Project fields (required)

- Title
- Type (Campus/Business)
- Profile image (required)
- Gallery images (optional, multiple)
- Video (optional)
- Short summary
- Full description
- Tech stack tags
- Featured toggle (optional)
- Confidential mode toggle (optional, recommended for Business projects)
- Status + isVisible + visibleFrom/visibleUntil
- Dates: createdAt, updatedAt, publishedAt
- SEO slug (auto-generated; editable optional)

Publish date display (Projects/Posts/Blogs/Ads)

- Frontend must display publish date on cards and detail pages (e.g., 'Published on 01 Mar 2026').
- If updated after publish, optionally show 'Updated on ...'.
- If scheduled, optionally show 'Scheduled for ...'.
- Admin has a toggle per item to show/hide date (default: show).

Project card UI (public)

- Card includes: profile image (or video overlay icon), type badge, publish date, title, summary, tech tags.
- Buttons: Request Project, Contact on WhatsApp (prefilled message), Save Project (Wishlist), View Details (direct link).
- Clicking anywhere on the card opens the project detail page in browser.

Project links (SEO friendly)

Each project must have an SEO-friendly URL using slug, e.g.
/projects/ai-based-ecommerce-platform

Media positioning controls (Admin adjustable)

- Admin can adjust media position for images/videos per project (and similarly for posts/blogs/ads/sections).
- Default must be a proper responsive layout (no stretching). Recommended default: object-fit: cover, centered.
- Supported controls: cover/contain, focus top/center/bottom, alignment left/center/right, media height (S/M/L/full), video ratio (16:9, 4:3, auto).

5. Project Requests (Lead Capture)

User flow

- From a project card or detail page, user can request the project via form or contact via WhatsApp.
- WhatsApp button opens a prefilled message including the project title and link.

Request form fields (minimum)

- Name
- Email
- Phone
- Budget range
- Preferred timeline (optional)
- Requirements / message
- Attachment upload (optional: PDF/DOC/images)

Admin request management

- Admin inbox for requests with filters (project, date range, status).
- Status pipeline: New -> Contacted -> In Progress -> Closed.
- Internal admin notes on each request.
- Export requests to CSV (optional PDF export).
- Soft delete + Recycle Bin support.

6. Collaboration Requests

Public availability

Anyone can request collaboration via a dedicated page (e.g., 'Collaborate With Us') and/or a footer CTA. This feature must be toggleable and schedulable by admin.

Collaboration form fields

- Full name
- Email
- Phone (optional)
- Country
- LinkedIn / Portfolio URL
- Collaboration type: Developer Partnership, Business Partnership, AI Research, Campus Team, Investment, Other
- Description / proposal
- File upload (optional proposal PDF)
- Consent checkbox for contact

Admin collaboration inbox

- View/filter by type, date, and status.
- Statuses: New, Reviewed, Accepted, Rejected, In Discussion.
- Admin notes.
- Optional: send response via email; WhatsApp shortcut.
- Export CSV; soft delete + Recycle Bin.

7. User Accounts & Profiles

User capabilities

- Register, login, logout (JWT).
- Maintain profile (basic details).
- Save projects to wishlist ('Add to Cart' concept implemented as Saved Projects).
- View saved projects and submitted requests in profile.
- View notifications and mark as read.
- Submit feedback with rating and images.
- Optional: add a payment card in profile (only store payment gateway token; never store raw card numbers).

Admin user management

- View user list and individual profiles.
- Block/unblock users.
- Soft delete users (restore via Recycle Bin).
- Access user activity logs (login/logout, wishlist actions, requests, feedback).

8. Feedback, Testimonials & Reviews

User feedback submission

- Rating (stars) + feedback text.
- Optional image uploads.
- Validation: required rating and message length rules.

Admin moderation

- Approve/hide feedback (spam control).
- Mark feedback as 'Featured' to display on landing page.
- Optional: reply to feedback.
- Soft delete + Recycle Bin.

9. Posts, Blogs, Ads, Promotions & Updates

Common requirements (applies to posts/blogs/ads/updates)

- Create/edit content in admin panel with statuses, visibility toggle, and scheduling window.
- Show publish date on cards and detail views (admin can toggle date display).
- Admin can preview before publish.
- Admin can adjust image/video position and size; default is clean responsive rendering.
- Soft delete + Recycle Bin.

Ads & Promotions

- Admin can create promotional banners or ads for landing page.
- Placement: hero, banner section, top bar, popup, or dedicated section.
- Optional CTA button (label + link).
- Scheduling for campaign periods (visibleFrom/visibleUntil).

Blog / Articles (optional but recommended)

- Blog list page + detail pages with SEO slugs (/blog/{slug}).
- Fields: title, featured image, publish date, author, tags, estimated read time, body content.
- Admin: schedule publish, preview, and media positioning.

10. Notifications & Promotions to Users

In-app notifications (required)

- Notification bell with unread counter for users.
- User can open notifications, mark as read, and optionally delete (soft delete).

Admin notification sending

- Admin can send: promotion, update, new project announcement, custom message.
- Targeting: all users, selected users, users who saved a specific project, users who requested a specific project.
- Optional deep-link attachment to a project/ad/post.
- Scheduling support (send now or schedule).

Email notifications (optional)

Optional: send email notifications for key events (new project published, major promotion). Use queued/async sending to avoid slowing the app.

11. Logging & Audit Trail

User activity logs (minimum)

- Login/logout (timestamp, IP, device/browser if available).
- Register.
- Profile updates.
- Wishlist saved/removed project.
- Submitted project request.
- Submitted feedback.

Admin audit logs (minimum)

- Create/update/delete (soft delete) for projects, posts, ads, updates, users, feedback.
- Publish/unpublish/schedule actions.
- Restore from recycle bin.
- Permanent delete from recycle bin.

Admin log viewer

- Filter by date range, user/admin, module, action type.
- Pagination and export to CSV.

12. Recycle Bin (Soft Delete & Recovery)

To guarantee no data loss, deleting content never removes it immediately. All deletes are soft deletes.

Soft delete fields (per table)

- deleted (boolean)
- deletedAt (datetime)
- deletedBy (admin/user id)
- deleteReason (optional)

Recycle Bin functions

- View deleted items by type (Projects, Ads/Posts, Updates, Feedback, Users, Requests, Page Sections, etc.).
- Restore item (returns to previous status/visibility).
- Permanently delete item (explicit confirmation dialog).
- Optional retention policy: auto-permanent delete after N days (configurable).
- When permanently deleting, also remove associated uploaded media to avoid storage bloat.

13. Preview Before Publish

Preview requirements

- Admin can preview any DRAFT/SCHEDULED item before publishing: home page, projects, posts/blogs, ads/promotions, updates.
- Preview must be token-protected and admin-only (JWT + expiring preview token).
- Preview must show: final layout, buttons, media positions, and publish date display.
- Device preview modes: Desktop, Tablet, Mobile breakpoints.

Example preview URLs

/preview/home?token=XXXX

/preview/projects/{slug}?token=XXXX

14. Admin Panel (Complete Functions)

Admin must be able to control everything via UI

- Dashboard: KPIs (users, projects, requests, feedback, posts/ads) + recent activity feed.
- Page Builder: edit front page and other pages via templates; reorder/hide/schedule; preview/publish.
- Front Page Edit Mode: inline editing controls (admin-only).
- Projects: CRUD + media upload + type (Campus/Business) + featured + confidentiality + status/visibility/scheduling + preview.
- Requests: view/manage pipeline statuses + notes + export.
- Collaboration: inbox + statuses + notes + export.
- Users: view/block/unblock/soft delete; view user logs.
- Feedback: approve/hide/feature + restore/delete.
- Posts/Blogs/Ads/Promotions/Updates: create/manage + scheduling + preview.
- Notifications: create/send/schedule; audience targeting; link attachments.
- Logs: user logs + admin audit logs with filters/export.
- Recycle Bin: restore/permanent delete across all modules.
- Settings: site branding, social links, WhatsApp settings, SEO defaults, footer content.

Admin access link

The Admin entry point must appear only as a footer link on the public site.

15. Settings, Branding & Social Links

Social media icons

- Header/footer social icons for: WhatsApp, Facebook, Instagram, LinkedIn, YouTube, GitHub (configurable).
- Admin can enable/disable each icon, set URL, and reorder icons.
- Icons open in new tab; only enabled icons are displayed.

Business configuration

- Site name, logo, favicon.
- Theme colors (optional light/dark mode).
- WhatsApp number + default prefilled message template.
- Footer content: address/email/phone, quick links, copyright text.
- SEO defaults: meta title/description templates; open graph image (optional).

16. Security, Validation & Non-Functional Requirements

Security

- JWT authentication; role-based authorization for ADMIN and USER.
- BCrypt password hashing; force admin password change on first login.
- Secure file uploads: validate file type/size, store safely, prevent path traversal.
- Prevent XSS/HTML injection: store structured content; never render raw unsafe HTML.
- Rate-limit sensitive endpoints (login, public forms) if possible.

Validation & UX quality

- Client-side + server-side validation for all forms (requests, collaboration, feedback, auth).
- Clear error messages and success confirmations.
- Performance: pagination for tables; lazy-load images; optimized media.
- Accessibility: readable contrast, keyboard navigation for forms, proper labels.

Responsiveness

- Public project grid: 3 cards desktop, 2 cards tablet, 1 card mobile.
- Admin panel: collapsible sidebar; tables must be usable on mobile (stacked rows or horizontal scroll).

17. Database Entities (Minimum Set)

Tables / entities required

- users
- roles (optional separate table; or role field on user)
- projects
- project_media
- project_requests
- collaboration_requests
- feedback
- notifications
- ads / posts / promotions (can be unified content table with type)
- blogs (optional but recommended) or unified content table
- pages
- page_sections (with content JSON and scheduling fields)
- settings
- logs (user logs + admin audit logs)
- saved_projects (wishlist)

Common columns (recommended for most tables)

- id (primary key)
- status, isVisible, visibleFrom, visibleUntil
- createdAt, updatedAt, publishedAt
- deleted, deletedAt, deletedBy
- createdBy, updatedBy (for admin-owned content)

End of specification